Markov Random Field Texture Models

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Abstract—We consider a texture to be a stochastic, possibly periodic, two-dimensional image field. A texture model is a mathematical procedure capable of producing and describing a textured image. We explore the use of Markov random fields as texture models. The binomial model, where each point in the texture has a binomial distribution with parameter controlled by its neighbors and "number of tries" equal to the number of gray levels, was taken to be the basic model for the analysis. A method of generating samples from the binomial model is given, followed by a theoretical and practical analysis of the method's convergence. Examples show how the parameters of the Markov random field control the strength and direction of the clustering in the image. The power of the binomial model to produce blurry, sharp, line-like, and blob-like textures is demonstrated. Natural texture samples were digitized and their parameters were estimated under the Markov random field model. A hypothesis test was used for an objective assessment of goodness-of-fit under the Markov random field model. Overall, microtextures fit the model well. The estimated parameters of the natural textures were used as input to the generation procedure. The synthetic microtextures closely resembled their real counterparts, while the regular and inhomogeneous textures did not.

Index Terms—Binomial model, goodness-of-fit, hypothesis test, image modeling, Markov random field, texture.

I. Introduction

THE subject of image modeling involves the construction of models or procedures for the specification of images. These models serve a dual role in that they can describe images that are observed and also can serve to generate synthetic images from the model parameters. We will be concerned with a specific type of image model, the class of texture models.

There are four important areas of image processing in which texture plays an important role: classification [25], [51], image segmentation [15], [40], [49], realism in computer graphics [5], [6], [9], [10], [13], [17], and image encoding [16], [39]. Besides being an intrinsic feature of realistic objects [32], [42], texture also gives important information on the depth and orientation of an object. Julesz [30], [31] considers the problem of generation of familiar textures an important one from both the theoretical and practical viewpoints. Understanding texture is also an essential part of understanding human vision [36], [37]. These considerations have led to an increased activity in the area of texture analysis and synthesis.

There is no universally accepted definition for texture. Part of the difficulty in giving a definition of texture is the extremely large number of attributes of texture that we would like to subsume under a definition. We consider a texture to be a stochastic, possibly periodic, two-dimensional image field. We mention a study by Tamura *et al.* [48] which attempted to find statistical features corresponding to the usual attributes of texture. The study delimited six attributes: coarseness, contrast, directionality, line-likeness, regularity, and roughness.

Most texture research can be characterized by the underlying assumptions made about the texture formation process. There are two major assumptions, and the choice of the assumption depends primarily on the type of textures to be considered in the study. The first assumption, which is called the placement rule viewpoint [43], [53], considers a texture to be composed of primitives. These primitives may be of varying or deterministic shape, such as circles, hexagons, or even dot patterns. Macrotextures have large primitives, whereas microtextures are composed of small primitives. These terms are relative to the image resolution [14]. The textured image is formed from the primitives by placement rules which specify how the primitives are oriented, both on the image field and with respect to each other. Examples of such textures include tilings of the plane, cellular structures such as tissue samples, and a picture of brick wall.

The second viewpoint regarding texture generation processes involves the stochastic assumption. The placement rule paradigm for textures may include a random aspect; in the stochastic point of view, however, we take a more extreme position and consider that the texture is a sample from a probability distribution on the image space. The image space is usually an $N \times N$ grid and the value at each grid point is a random variable in the range $\{0, 1, \cdots, G-1\}$. Textures such as sand, grass, and water are not appropriately described by a placement model. The key feature of these images is that the primitives are very random in shape and cannot be easily described.

In this paper we will explore the use of a Markov random field model for the generation and analysis of textured images. The goal of the research is to produce a texture analysis and synthesis system which will take as input a texture, analyze its parameters according to the Markov random field model, and then generate a textured image that both resembles the input texture visually and matches it closely from a statistical point of view. This can be considered a kind of Turing test for image generation [50], in that the proof of the viability of the system will be to produce textures that cannot be distinguished by humans from their real counterparts. We will not perform a rigorous psychological study of the correspondence, but will concentrate on the statistical evaluation of the goodness-of-fit of the observed texture and the generated texture.

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II. Texture Models

By a model of a texture, we mean a mathematical process which creates or describes the textured image. The primary goal of texture modeling is the description of real textures. A secondary goal of texture modeling is classification of textures. The numerical parameters of the model can be used as features to classify the texture. There is a distinction between model-based studies and attempts to find good features for the classification of textures. In a model-based environment, we have the capability to produce, for example, textures that match observed textures. In a feature-based texture analysis, the textural features are measured without an ideal or representative texture in mind. A substantial portion of texture research has been done at the feature-based level. A survey of the well-known and commonly used texture features appears in a recent paper [26].

Our interest here is in texture models which take the stochastic process approach. Ideally, we would like to find a stochastic process that is physically meaningful and related to the texture which we are modeling. One can use more prior information about the texture in a model-based approach than in the feature-based approach. In the stochastic process approach, brightness levels or pixel gray values are the random variables. The level $X(i,j)$ at some point (i,j) is not independent of the levels at other points in the image. In fact our principal concern is about the correlations between the $\{X(i,j)\}$.

The models which have been used to generate and represent textures include:

1) time series models [34]
2) fractals [35]
3) random mosaic models [2], [39], [44], [45]
4) mathematical morphology [46]
5) syntactic methods [18], [33]
6) linear models [15], [16].

III. Markov Random Fields

The brightness level at a point in an image is highly dependent on the brightness levels of neighboring points unless the image is simply random noise. In this section, we explain a precise model of this dependence, called the Markov random field. The notion of near-neighbor dependence is all-pervasive in image processing. Focusing directly on this property is a promising approach to the overall problem of microtextures.

The study of Markov random fields has had a long history, beginning with Ising's 1925 thesis [29] on ferromagnetism. Although it did not prove to be a realistic model for magnetic domains, it is approximately correct for phase-separated alloys, idealized gases, and some crystals [41]. The model has traditionally been applied to the case of either Gaussian or binary variables on a lattice. Besag [4] allows a natural extension to the case of variables that have integer ranges, either bounded or unbounded. These extensions, coupled with estimation procedures, permit the application of the Markov random field to texture modeling.

The Markov random field model has been briefly investigated by Hassner and Sklansky [27]. Their work was limited to an exposition of the equivalence between the Gibbs field and Markov random field expressions for the conditional probability distributions (see Spitzer [47] for the proof) and generation of a few examples of textures. Moreover, they limited their attention to the binary case.

An alternative to the full two-dimensional Markov random field for images is to traverse the lattice along a scan line and provide a direct analog to the usual Markov chain [1]. Connors and Harlow [11] generated streaky line textures according to a simple Markov chain that ignores the correlations between pixels in neighboring rows. Haralick and Yokoyama [52] generated essentially one-dimensional textures using scans, but provided some correlations between neighboring rows by considering changes in the features computed from the co-occurrence matrices.

A. Definitions

Our exposition follows Besag [4] and Bartlett [3]. Let $X(i,j)$ denote the brightness level at a point (i,j) on the $N \times N$ lattice L. We simplify the labeling of the $X(i,j)$ to be $X(i)$, $i = 1, 2, \cdots, M$ where $M = N^2$.

Definition 1: Let L be a lattice. A *coloring of* L (or a *coloring of* L *with* G *levels*) denoted $\boldsymbol{X}$ is a function from the points of L to the set $\{0, 1, \cdots, G-1\}$. The notation $\mathbf{0}$ denotes the function that assigns each point of the lattice to 0.

Definition 2: The point j is said to be a neighbor of the point i if

$$p(X(i)|X(1), X(2), \cdots, X(i-1), X(i+2), \cdots, X(M))$$

depends on $X(j)$.

Note that Definition 2 does not imply that the neighbors of a point are necessarily close in terms of distance, although this is the usual case. Now we can give the definition of a Markov random field.

Definition 3: A *Markov random field* is a joint probability density on the set of all possible colorings $\boldsymbol{X}$ of the lattice L subject to the following conditions.

1) *Positivity:* $p(\boldsymbol{X}) > 0$ for all $\boldsymbol{X}$.
2) *Markovianity:*

$$p(X(i)|\text{all point in the lattice except } i) = p(X(i)|\text{neighbors of } i)$$

3) *Homogeneity:* $p(X(i)|\text{neighbors of } i)$ depends only on the configuration of neighbors and is translation invariant (with respect to translates with the same neighborhood configuration).

The Hammersley-Clifford theorem [4] delimits the functions admissible as conditional probability distributions at each point of the lattice and provides a connection between the purely graph-theoretic relationships on a lattice with the algebraic form of the density function. The general formulation of distributions satisfying the above three conditions is given by Besag [4]. We limit our attention to the case where the probability of a point $X(i,j)$ having gray level k is binomial, with parameter determined by its neighbors. This is Besag's formulation of the *autobinomial* distribution.

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B. Order Dependence and Anisotropy

In most cases, we are interested in the models where the point i is a neighbor of the point j if i is close to j.

The probability $p(X = k|\text{neighbors})$ is binomial with parameter $\theta(T)$ and number of tries $G - 1$ where G is the number of gray levels. The value of T is given in (1)-(5) for models of various orders. The $b(i, k)$ are the parameters of the model and are 0 for all i larger than the order. The value of a indirectly controls the lattice process. We have limited our attention to case of a maximum fourth-order dependence since higher order parameters cannot be estimated from 64 X 64 images:

$$\theta = \frac{\exp(T)}{1 + \exp(T)} \tag{1}$$

where a first-order model has the form for T:

$$T = a + b(1, 1)(t + t') + b(1, 2)(u + u'). \tag{2}$$

A second-order model has a T of the form

$$T = a + b(1, 1)(t + t') + b(1, 2)(u + u') + b(2, 1)(v + v') + b(2, 2)(z + z'). \tag{3}$$

A third-order model takes the form

$$T = a + b(1, 1)(t + t') + b(1, 2)(u + u') + b(2, 1)(v + v') + b(2, 2)(z + z') + b(3, 1)(m + m') + b(3, 2)(1 + 1'). \tag{4}$$

Finally, a fourth-order model is obtained by adding an additional term of the form

$$b(4, 1)(o1 + o1' + o2 + o2') + b(4, 2)(q1 + q1' + q2 + q2') \tag{5}$$

to the form for the third-order T. Additional high-order terms can be obtained by extending the orders in a similar way beyond those in Fig. 1.

The formal definition of order can now be given.

Definition 4: The *order* of a Markov random field process on a lattice is the largest value of i such that $b(i, 1)$ or $b(i, 2)$ is nonzero.

Definition 5: A Markov random field is *isotropic at order i* if $b(i, 1) = b(i, 2)$. Otherwise, it is said to be *anisotropic at order i*. The notation $b(i, \cdot)$ implies isotropy at order i and signifies the common value of $b(i, 1) = b(i, 2)$.

The notion of anisotropy agrees with our intuitive notion of directionality in textures. For example, positive values of the clustering parameter $b(1, 1)$ cause clustering in the horizontal direction, whereas $b(1, 2)$ controls clustering in the vertical direction. The $b(2, 1)$ and $b(2, 2)$ control clustering in the diagonal directions. Positive values of these parameters cause attraction; negative values result in repulsion, or a checkerboard effect. More complex types of anisotropy and nonlinear terms can be put into (2)-(5). Cross [12] discusses other models.

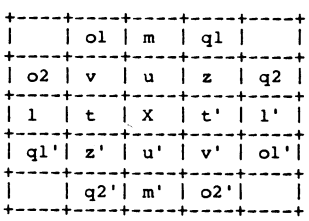


Fig. 1. Neighbors of the point X.

The binary case, where the point variables have range $\{0, 1\}$, is a special case of the binomial model. We obtain the conditional probability of x by

$$p(X = x|T) = \frac{\exp(xT)}{1 + \exp(T)}. \tag{6}$$

When dealing with a finite rectangular lattice and using the definitions of neighbor and order above, points on the edge of the lattice have fewer neighbors than the interior points. We compensate for this by assuming that the lattice has a periodic or torus structure. This means that the left edge is connected to the right edge and the upper edge is connected to the lower edge.

IV. Simulation of Markov Random Fields

In order to generate textures that are the visual representation of Markov random fields, we need a procedure that yields a sample from a Markov random field with given parameters. Fortunately, such procedures exist and have been used extensively in physics to investigate the properties of two- and three-dimensional Ising lattice [7], [22], [20]. Hassner and Sklansky use a procedure that seems to be similar to the classical ones with the addition of a preprocessing step that seems to eliminate some unlikely neighborhood configurations [28]. Simulation is usually needed to estimate many of the properties of Markov random fields since the analytical calculations are, for the most part unsatisfactory [21].

The required theory for the simulation of Markov random fields comes from the theory of discrete, finite-state Markov chains. We want a Markov chain whose states are the set of colorings $\{X\}$ with limiting distribution $\{p(X)\}$. We can sample such a chain and observe colorings $\{X\}$ with frequency given by $\{p(X)\}$. Theorems are available which do the following.

1) Give sufficient conditions for a Markov chain to have a unique limiting distribution [19, p. 393]. These conditions are met in our case.

2) Show how to convert a relatively arbitrary symmetric Markov chain to one with specified limiting distribution $\{q(j)\}$. The key feature of this theorem is the fact that we need only know the ratios $\{q(j)/q(i)\}$ in order to obtain the desired chain [24].

Theorem: Consider a symmetric, aperiodic, irreducible Markov chain with transition matrix P^*. Let $\{q(j)\}$ be a set of positive numbers with sum 1. Then the Markov chain with transition matrix P has limit distribution $\{q(j)\}$, where P is

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```
while not STABLE do
     begin
          choose sites X(1),X(2) with X(1) <> X(2);
          r := P(Y)/P(X);
          if r => 1
               then switch X(1),X(2)
               else
                    begin
                         u:=uniform random on [0,1];
                         if r > u
                            then switch X(1),X(2)
                            else retain X
                         end
     end;
```

Fig. 2. Algorithm for generating Markov random field with joint probability function $p(\boldsymbol{X})$. The coloring $\boldsymbol{Y}$ is obtained from the coloring $\boldsymbol{X}$ by switching the values of the points $X(1)$ and $X(2)$. Variables with underbars appear boldface in text.

defined by

$$p(i,j) = \begin{cases} p^*(i,j)q(j)/q(i) & \text{if } q(i) > q(j) \\ p^*(i,j) & \text{if } q(j) \geqslant q(i) \end{cases}$$

$$p(i,i) = p^*(i,i) + \sum_j{}' p^*(i,j)(1 - q(j)/q(i))$$

where the prime on the summation means summation over all indices with j with $q(j)/q(i)$ less than 1.

3) Allow us to calculate the set of ratios $\{q(j)q/(i)\} = \{p(\boldsymbol{Y})/p(\boldsymbol{X})\}$ from the conditional distribution of a Markov random field without explicit calculation of the set of $\{p(\boldsymbol{X})\}$ [4].

Theorem: Let $\boldsymbol{X}$ and $\boldsymbol{Y}$ be two colorings of the Markov random field lattice L. Then

$$\frac{p(\boldsymbol{Y})}{p(\boldsymbol{X})} = \prod_{i=1}^{M} \frac{p(X(i) = y(i) | X(1), X(2), \cdots, X(i-1), Y(i+1), \cdots, Y(N))}{p(X(i) = x(i) | X(1), X(2), \cdots, X(i-1), Y(i+1), \cdots, Y(N))}.$$

M attempted exchanges or switches to constitute one *iteration*. Notice that this ignores attempted exchanges between pixels of the same color. It was observed experimentally that in ten iterations or less, either the number of changes drops to 1 percent of M or the measured parameters match the input parameters to within about 5 percent. These guidelines define the variable "STABLE" in Fig. 2. On a PDP-11/34 computer, the time required for one iteration on a 64 X 64 image was 2-3 min depending on the number of gray levels.

We give an example of the convergence for a specific case. The purpose of this example is not to prove convergence, which is a consequence of the mathematical results discussed at the beginning of this section, but rather to give some indication of the convergence rate. We want to simulate a first-order binary Markov random field with parameters

$$a = -2$$

$$b(1, \cdot) = 1$$

on a 128 X 128 lattice. The estimated parameters $b(1, \cdot)$ are shown plotted against number of iterations in Fig. 3. The graph flattens rather quickly and stays rather within 0.05 of the intended value of 1.0 for $b(1, \cdot)$. The parameter estimate which is plotted in Fig. 4 is the average of two estimates, performed on two different codings, as explained in Section V-A. The two estimates agree within about 12 percent. Fig. 4 shows the number of changes observed per 256 attempted exchanges, as the estimates were made every 1/64 iteration. Thus, although in an iteration of M attempted exchanges we are still observing nearly $M/2$ changes, the chain is near equilibrium, as the observed model parameters are within a small tolerance of the intended parameters.

A. Texture Generation Procedure

In general, we are interested in textures with the same number of pixels at each gray level. This is done by starting with an image that is generated by coloring the point (i,j) with level k, where k is chosen with equal probability from the set $\{0, 1, 2, \cdots, G-1\}$. The convergence to the limit distribution is unaffected by the choice of initial configuration; only the rate at which equilibrium is reached depends on the choice of the initial configuration. Given a state (i.e., coloring) $\boldsymbol{X}$, we choose the next state $\boldsymbol{Y}$ to be the same as $\boldsymbol{X}$ except that the gray values of two randomly selected points are interchanged. The algorithm is diagrammed in Fig. 2. The algorithm was used by Flinn [21] and was invented by Metropolis *et al.* [38].

B. Convergence Properties

A theorem of Hammersley [24] guarantees that the application of the algorithm in Fig. 2 will eventually result in a lattice with the desired distribution. The practical question is how long this will take.

We first need to define a time-dimension for the simulation. Suppose that the lattice is $N \times N$ and let $M = N^2$. We consider

C. Examples of Textures

We present some examples of textures generated according to various settings of Markov random field parameters. These images are representative of the kind of results that can be

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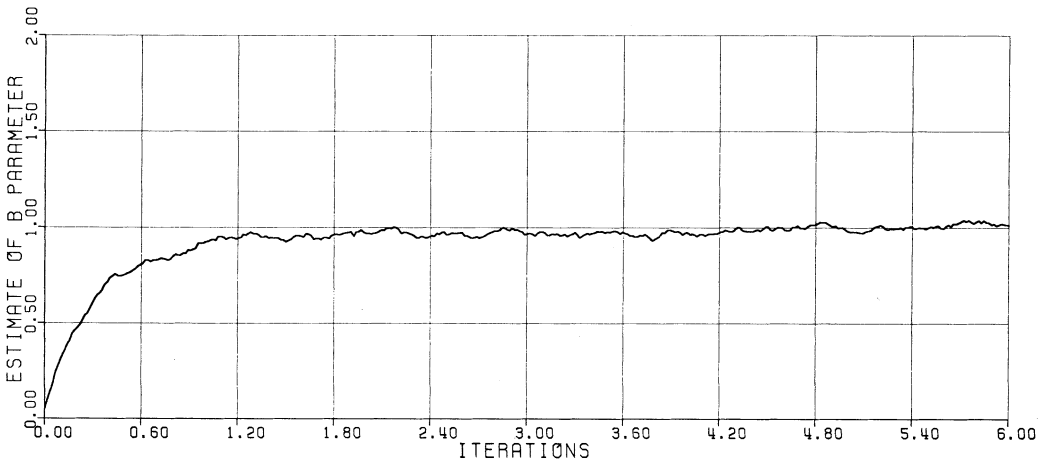


Fig. 3. Convergence of $b(1, \cdot)$.

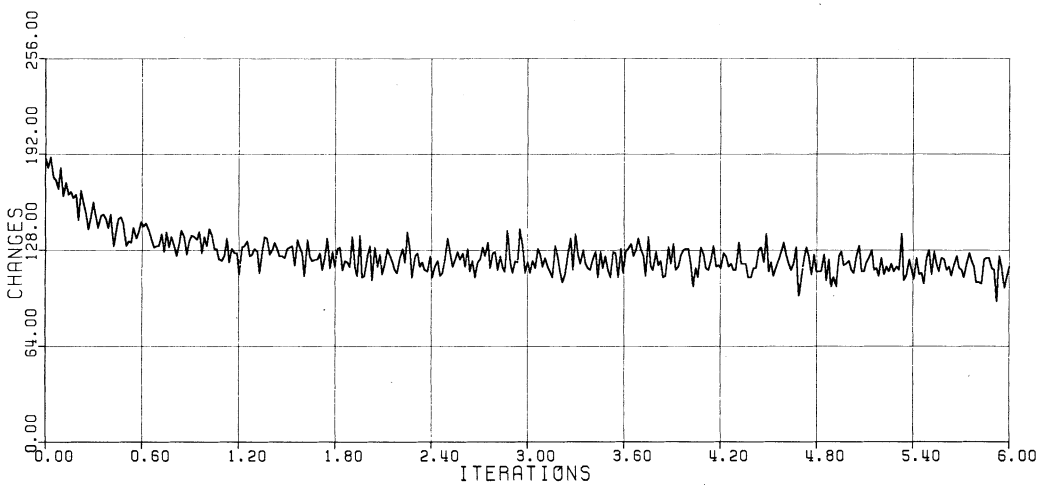


Fig. 4. Behavior of number of exchanges.

achieved but are not necessarily attempts to imitate real textures. They should rather be considered to be an "alphabet" of Markov random field textures. In Section V, we exhibit generated textures matching observed textures.

1) *Clustering Effects:* Fig. 5 shows a series of 64 X 64 binary textures generated according to an isotropic first-order model. Fig. 5(a) represents "noise," i.e., the probability of a pixel being black or white is 0.5, independently of all other pixels. In Markov random field terms,

$$a = b(1, \cdot) = 0.$$

The value of $b(1, \cdot)$ is increased from 0 to Fig. 5(a) to 3.0 in Fig. 5(h). The increase in clustering is clearly visible.

2) *Anisotropic Effects:* Fig. 6 shows extreme anisotropy in first-order and second-order model on a 64 X 64 lattice. The quantity $b(1, 1)$ controls the horizontal clustering; $b(1, 2)$ causes vertical clustering. In Fig. 6(a), large positive values of $b(1, 2)$ and large negative values of $b(1, 1)$ result in "clean" vertical lines. Contrast this with Fig. 6(b), which has thickened and noisy horizontal lines because of the small positive value of the vertical clustering parameters. The decidedly diagonal effect of Fig. 6(c) results from the use of a second-order structure. The clustering *i* the NW-SE direction is pronounced since the parameter in this direction is 1.9 and the parameters in all other directions are quite small.

3) *Ordered Patterns:* Many of the applications of the Ising model involve studying the checkerboard-like patterns obtained with negative clustering parameters [41]. This is illustrated by Fig. 7, where the most likely configuration is a black pixel surrounded by four white pixels or vice versa.

4) *Attraction-Repulsion Effects:* An attraction-repulsion process involves having low-order parameters positive, resulting, in clustering but high-order parameters negative in order to inhibit the growth of clusters. If high-order parameters were also positive, large clusters would result, whereas negative high-order parameters yield small clusters. Fig. 8 shows the effect of anisotropic clustering with inhibition. Fig. 8(b) contains longer horizontal and vertical lines than Fig. 8(a) because of the larger values of the first-order clustering parameters $b(1, 1)$ and $b(1, 2)$. Fig. 9 shows two isotropic attraction-repulsion textures. Cluster sizes are small here because of the high-order inhibition.

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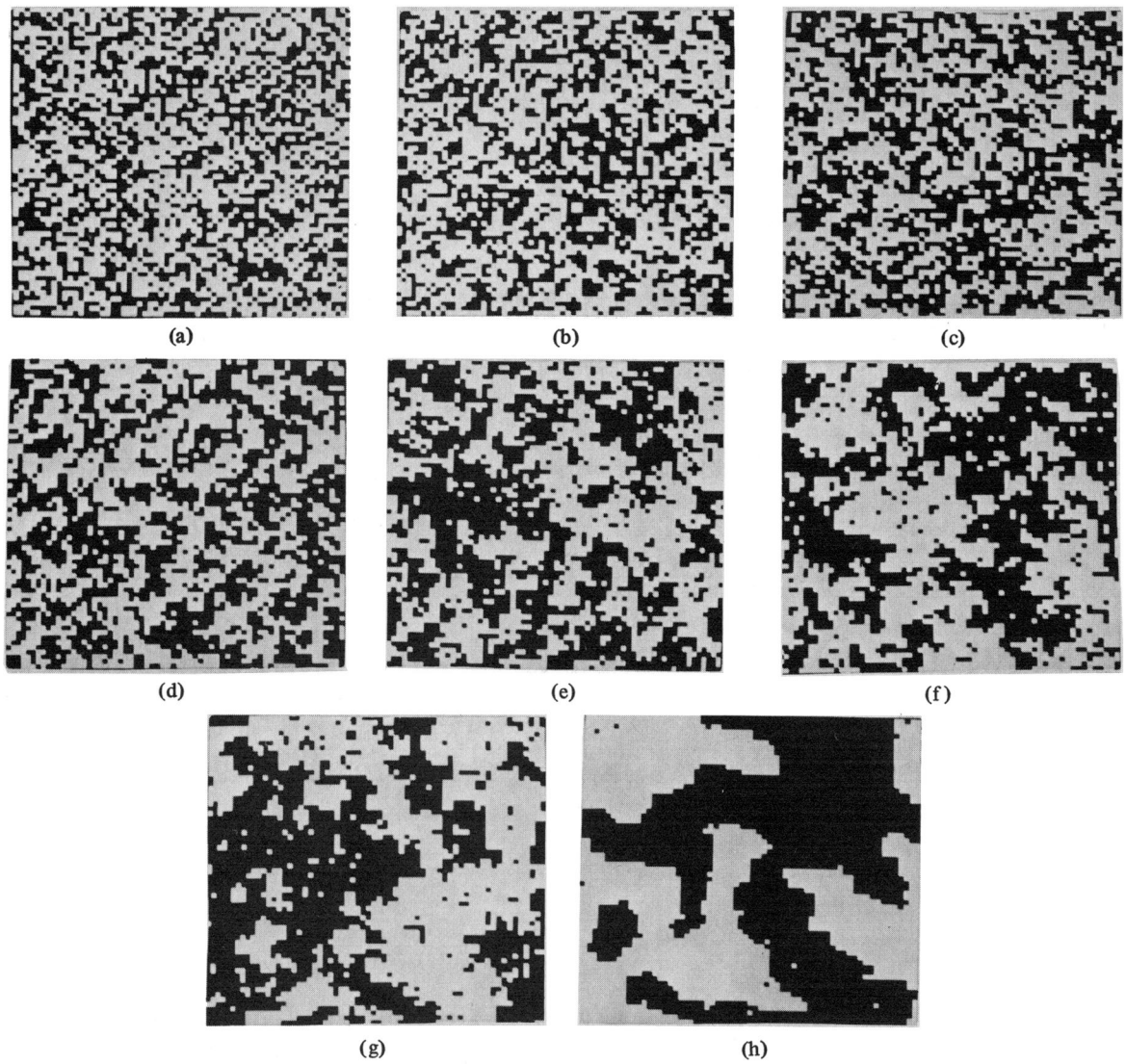


Fig. 5. Isotropic first-order textures. The $b(1, \cdot)$ parameters are: (a) 0.0, (b) 0.50, (c) 0.75, (d) 1.1, (e) 1.26, (f) 1.52, (g) 1.79, (h) 3.0. In all cases, the a parameter is $-2b(1, \cdot)$.

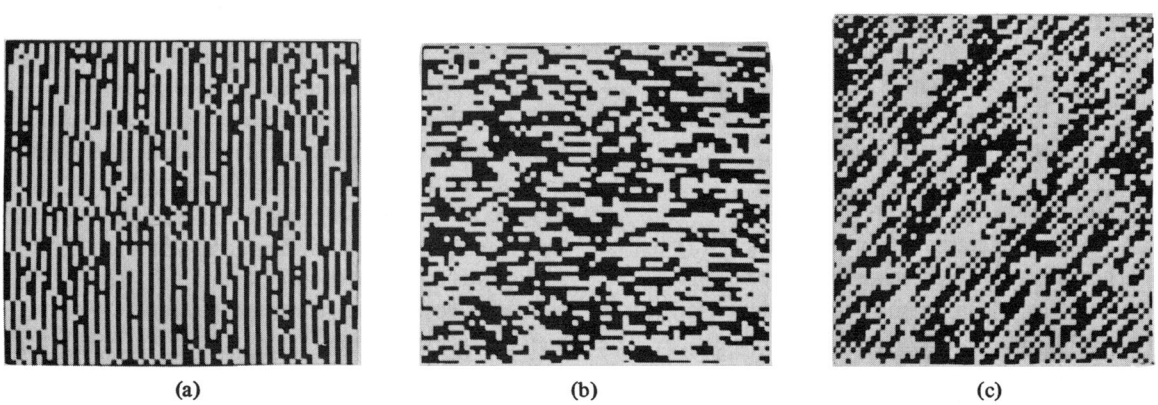


Fig. 6. Anisotopic line textures. The parameters are (a) $a = -0.26$, $b(1, 1) = -2$, $b(1, 2) = 2.1$, $b(2, 1) = 0.13$, $b(2, 2) = 0.015$. (b) $a = -2.04$, $b(1, 1) = 1.93$, $b(1, 2) = 0.16$, $b(2, 1) = 0.07$, $b(2, 2) = 0.02$. (c) $a = -1.9$, $b(1, 1) = -0.1$, $b(1, 2) = 0.1$, $b(2, 1) = 1.9$, $b(2, 2) = -0.075$.

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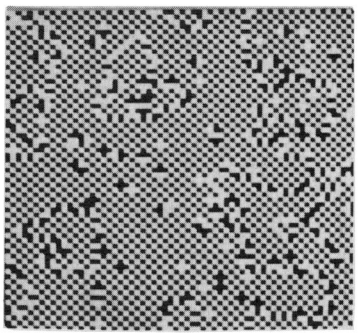
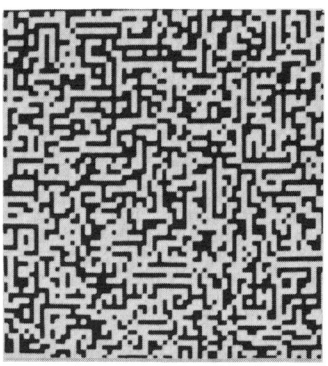
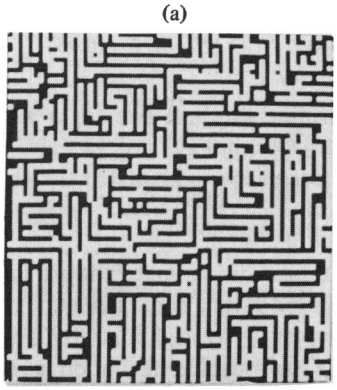


Fig. 7. Ordered pattern. The parameters are $a = 5.09$, $b(1, 1) = -2.25$, $b(1, 2) = -2.16$.



(a)



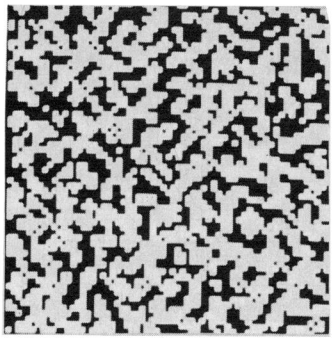
(b)

Fig. 8. Diagonal inhibition textures. The parameters are (a) $a = 2.19$, $b(1, 1) = -0.088$, $b(1, 2) = -0.009$, $b(2, 1) = -1$, $b(2, 2) = -1$. (b) $a = 0.16$, $b(1, 1) = 2.06$, $b(1, 2) = 2.05$, $b(1, 2) = -2.03$, $b(2, 2) = -2.10$.

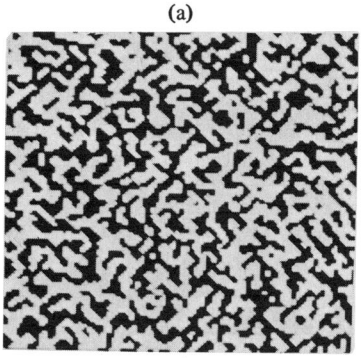
5) *Multiple Gray Scale Textures:* We now turn our attention to the binomial model. Fig. 10 shows a 4-gray level picture with considerable clustering. Fig. 10(a) is isotropic and first-order, whereas Fig. 10(b) is second-order anisotropic with diagonal clustering. Fig. 11(a) and (b) represents typical multiple gray scale pictures with isotropic clustering.

Fig. 12 shows a pattern similar to Fig. 6, but with 32 gray levels. The resemblance to wood-grain is apparent. Fig. 13(a) and (b) shows the result of attraction-repulsion processes with multiple gray levels. Fig. 13(a) has the appearance of reticulated photographic film due to strong third- and fourth-order inhibition. The diagonality in Fig. 13(b) is a result of strong repulsion in some directions and clustering in others.

It should be noted that many of these images appear blurry

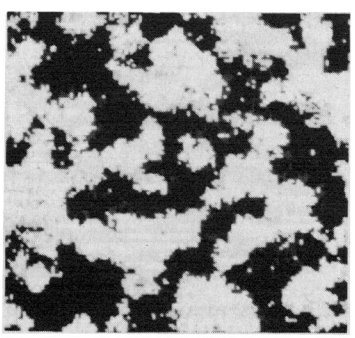


(a)

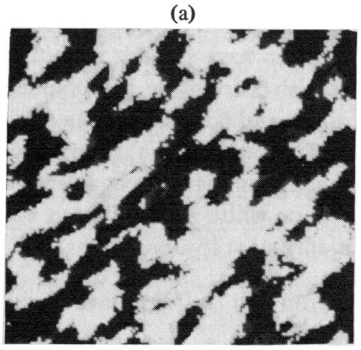


(b)

Fig. 9. Isotropic inhibition textures. The parameter values are (a) $a = -0.97$, $b(1, \cdot) = 0.94$, $b(2, \cdot) = 0.94$, $b(3, \cdot) = -0.42$, $b(4, \cdot) = -0.49$. (b) $a = -4.6$, $b(1, \cdot) = 2.62$, $b(2, \cdot) = 2.17$, $b(3, \cdot) = -0.78$, $b(4, \cdot) = -0.85$.



(a)



(b)

Fig. 10. Clustered textures with 4 gray levels. The parameters are (a) $a = -2$, $b(1, 2) = 1.0$. (b) $a = -2$, $b(1, \cdot) = 1.0$, $b(2, 1) = 1.0$, $b(2, 2) = 0.0$.

and out of focus. This effect is not due to the reproduction process but is intrinsic to the binomial model. If there is no inhibition (via negative high-order parameters), then the bi-

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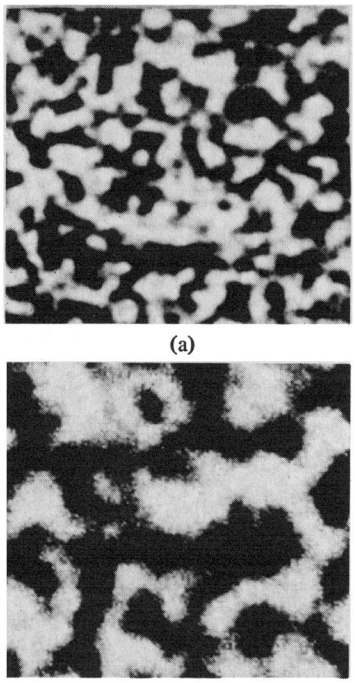


Fig. 11. Clustered textures with 16 and 32 gray levels. The parameters are (a) 16 levels, $a = -2.0$, $b(1, \cdot) = b(2, \cdot) = b(3, \cdot) = b(4, \cdot) = 0.05$. (b) 32 levels, $a = -2.0$, $b(1, \cdot) = b(2, \cdot) = b(3, \cdot) = b(4, \cdot) = 0.05$.

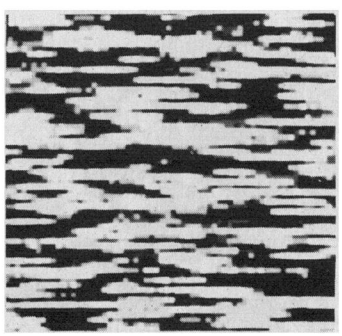


Fig. 12. Horizontal texture with 16 gray levels. The parameters are $a = -2.0$, $b(1, 1) = 0.08$, $b(1, 2) = 1.0$.

nomial model tends to have smooth transitions from black to white. The binomial distribution is unimodal and, as a consequence, values above and below the mean gray value are highly probable also. This results in a tapering of the gray scale around maxima and minima. Such a tapering as one moves away from black or white points has an effect similar to a neighborhood averaging or low-pass filter.

V. Modeling of Natural Textures

In previous sections, we have discussed the probabilistic structure of Markov random fields and have shown how samples from Markov random fields can be generated. We now implement a statistical measure of the correspondence between an observed texture and a texture model. No prior study has performed this kind of evaluation. All prior studies in texture modeling have considered a model adequate if its parameters yielded good classification in pattern recognition experiments or if it was found to be the best-fitting among a number of models tested. For example, Deguchi and Morishita [15] determine the best size for a neighborhood in an autoregressive scheme, but do not give any overall guidelines on when the autoregressive scheme fits the observed texture.

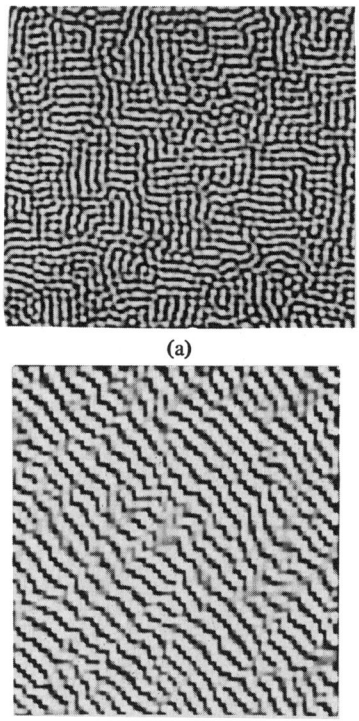


Fig. 13. Attraction–repulsion textures with 16 gray levels. The parameters are (a) $a = -2$, $b(1, \cdot) = b(2, \cdot) = 0.2$, $b(3, \cdot) = b(4, \cdot) = -0.1$. (b) $a = -2$, $b(1, \cdot) = 0.2$, $b(2, 1) = -2.2$, $b(2, 2) = 0.2$, $b(3, 1) = -0.5$, $b(3, 2) = 0.05$, $b(4, 1) = -0.05$, $b(4, 2) = 0.05$.

A. Estimation of Parameters

The technique used to estimate the parameters $\{b(j, k)\}$ is maximum likelihood estimation. Let $p(x|\cdot)$ denote the conditional probability $p(X = x|\text{neighbors of } X)$, where X is a point of the lattice L. The usual log likelihood is given by

$$1 = \sum_{X} \ln(p(X|\cdot)) \tag{7}$$

where the summation in (7) extends over all points of the lattice.

The summands in (7) are not independent. Instead of forming 1 as a sum over all points of the lattice, the lattice is partitioned into disjoint sets of points called *codings*. Each coding is chosen so that its points are independent. This can be done by adequately spacing the X points so that if $X(i)$ and $X(j)$ are two points in a coding, then $X(i)$ is not a neighbor of $X(j)$ in the Markov random field sense.

A first-order process requires at least two codings for estimation purposes. A second-order process requires spacing so that 3×3 neighborhoods do not interfere. This yields four codings. Nine codings are needed for third- and fourth-order processes. These codings are shown in Figs. 14, 15, and 16.

The actual estimation procedure is straightforward. Let $1(i)$ be the log likelihood for the ith coding obtained by extending the summation only over those points which are in coding i. In (7), $p(X|\cdot)$ depends on the order of the Markov

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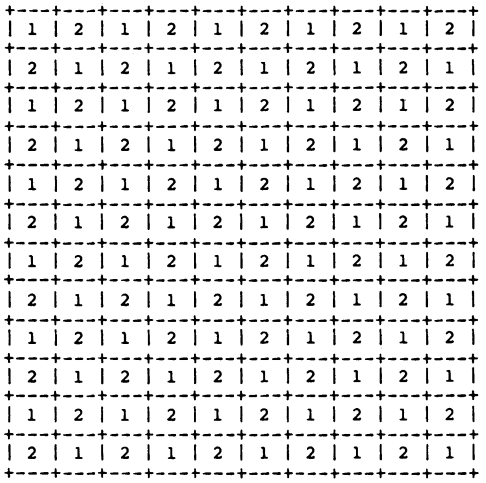


Fig. 14. First-order codings.

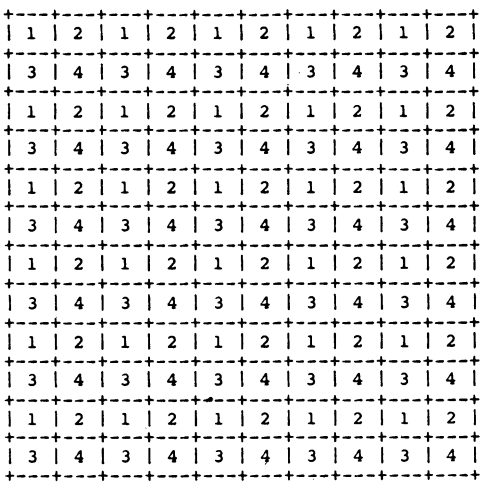


Fig. 15. Second-order codings.

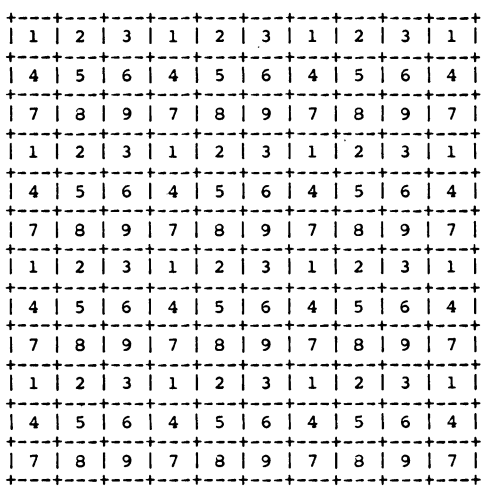


Fig. 16. Third- and fourth-order codings.

random field whose parameters are being estimated. An estimate of the parameter vector for the ith coding is obtained by maximizing $1(i)$. Our final estimate of the parameters is the average value over all the codings. Besag [4] mentions some

TABLE I
EXAMPLE OF A CHI-SQUARE TEST FOR A BINARY 64 × 64 TEXTURE. A FIRST-ORDER ANISOTROPIC SCHEME WAS FITTED. THE RESULTS OF CODING NUMBER 1 ARE SHOWN ABOVE. THE CHI-SQUARE VALUE IS 12.30957, ON 6*df*.

u + u'	v + v'	X = 0	X = 1
0	0	625(621.)	5(9.)
0	1	240(244.)	20(16.)
0	2	14(17.)	8(5.)
1	0	79(80.)	19(18.)
1	1	93(91.)	85(87.)
1	2	21(14.)	52(59.)
2	0	4(3.)	8(9.)
2	1	12(17.)	244(239.)
2	2	8(8.)	512(512.)

doubt about the efficiency of this procedure. We have noted little variation in the estimates of textures over various codings. For example, on the 16 binary brick textures estimated as an anisotropic first-order field, let $b(1, 1)$ and $b(1, 1)'$ denote the estimates of the horizontal parameters while $b(1, 2)$ and $b(1, 2)'$ denote the estimates of the vertical parameters on the two codings. Then over the 16 samples we observed that

$$0.24 = \sum |b(1, 1) - b(1, 1)'|$$

and

$$0.09 = \sum |b(1, 2) - b(1, 2)'|.$$

This represents about a 13 percent average difference for the horizontal parameter (1.82) and 3 percent average difference for the vertical parameter (2.75).

B. Hypothesis Testing

Note that we really have only one sample from the unknown distribution $p(X)$ on the set of colorings of the lattice L. From the conditional probability point of view, each observed configuration of neighbors and the value of the center point X is a sample. In this sense, we have $M = N^2/k$ samples of the conditional density $p(X|\cdot)$ for the $N \times N$ lattice 1, where k is the number of codings. We can thus perform a chi-square test of the fit between the expected frequencies for each center pixel and the observed frequency. The expected frequencies are computed using the estimated parameters with an appropriate reduction in degrees of freedom. The null hypothesis is as follows.

H_0: The texture is a sample from a Markov random field with the estimated parameters set $\{a, b(j, k)\}$

while the alternative hypothesis is simply the negation of H_0.

Table I shows an example of a chi-square test for a first-order, anisotropic binary lattice. The entries in the table are of the form "observed (expected)" and the expected entry is computed using the conditional probability distribution with the estimated parameters ($a = -4.26$, $b(1, 1) = 2.70$, and $b(1, 2) = 1.5$). The degrees of freedom are computed by means

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of the formula

$$df = (G - 1) \cdot N_C - E$$

where G is the number of gray levels, N_C is the number of neighborhood configurations (in Table I, each neighborhood configuration is a row), and E is the number of estimated parameters. In the example of Table I, we have $(2 - 1) \cdot 9 - 3 = 6$ df. We also use the convention of having at least one expected observation per cell. This results in a reduction in the number of cells and degrees of freedom.

Since we are performing a number of tests on the same data (one on each of k codings), there is a great likelihood of having the hypothesis of a fit to a Markov random field scheme accepted on some codings and rejected on others. The results of the tests are not independent. If they were independent, and we performed k tests at a level α, then the probability of no rejections would be $(1 - \alpha)^k$. A conservative strategy is to take the overall significance level of the test to be kp, where p is the level at which the most significant result was obtained over k codings [4].

The study of natural textures was based on 12 pictures from the Brodatz texture album [8]. The plates used are given in Tables II. Each plate was photographed on 35 mm film to form 24 X 36 mm slides. The slides were digitized in such a way that the small dimension occupied slightly over 256 pixels of the 480 X 640 image of a Spatial Data Eyecom system. The 256 X 256 images were split into 16 nonoverlapping 64 X 64 subimages and also into four 128 X 128 nonoverlapping subimages. The gray scale of each subimage was reduced from 256 gray levels to both two and eight gray levels using equal probability quantizing [25].

TABLE II
TEXTURES USED IN THE STUDY. THE PLATE NUMBERS REFER TO THE BRODATZ TEXTURE ALBUM [8].

Name	Plate Number
Brick Wall	D94
Ceiling Tile	D86
Pressed Cork	D4
Calf Fur	D93
Grass Lawn	D9
Handmade Paper	D57
Pebbles	D31
Beach Sand	D29
Straw Screening	D49
Water	D38
Wood Grain (1)	D69
Wood Grain (2)	D70

TABLE III
BEST FITTING BINARY TEXTURE MODELS. IN THE TABLE BELOW, THE "*I*" INDICATES THAT THE FEWEST REJECTIONS WERE OBTAINED BY USING ISOTROPIC ESTIMATION, WHILE "*A*" INDICATES THAT THE BEST RESULTS WERE OBTAINED USING ANISOTROPIC PARAMETERS. THE SYMBOL "—" IN THE SECOND-ORDER COLUMN SIGNIFIES THAT THE BEST RESULTS WERE OBTAINED USING A FIRST-ORDER MODEL.

Texture Name	First Order	Second Order	Acceptances at 5%
Bricks	I	---	16
Ceiling Tile	I	---	16
Cork	I	A	14
Fur	I	I	12
Grass	A	---	16
Paper	I	---	15
Pebbles	I	---	14
Sand	A	A	14
Screen	A	A	5
Water	A	A	14
Wood (1).	A	A	14
Wood (2)	A	A	14

C. Evaluation of the Fit

The textures were evaluated for their fit to various models. The purpose of this analysis is twofold. First, we need to validate that the Markov random field model is generally applicable to textured images. The second objective is to formulate some general guidelines on how to choose an appropriate model, in terms of order and the degree of anisotropy and isotropy, to generate specific types of textures.

1) *Binary Texture Results:* Except for the screen texture, some first- or second-order model was able to give at least ten acceptances, under the previously mentioned conservative decision rule, for each texture sample. Detailed analysis of the screen texture samples showed very few distinct neighborhood configurations. This is a consequence of its regularity. The few neighborhood configurations dominate the computation of the Markov random field parameters. However, the low frequency neighbor probabilities are not properly controlled by these parameters, resulting in large chi-square values. Essentially, the histogram is so skewed toward these configurations that the positivity condition of Section III-A is nearly violated.

A number of models were tried out for testing binary textures. A test against a first-order model results in two hypothesis tests, one for each coding. There are 16 subimages of each texture, and each may be rejected on zero, one, or two codings at the five percent level. The number of conservative acceptances, of which there are a maximum of sixteen possible, is recorded in Table III. Besag [4] makes the point that one cannot compare the fit of a second-order model to the fit of a first-order model unless the same coding scheme is used in both cases. Table III gives the best results on second-order coding for each of the binary textures. The general good fit of the first-order model should be reconciled with the fact that there are only two codings rather than the four for a second-order scheme, which means that fewer rejections are likely.

We have limited our attention to the case of 64 X 64 textures. Although third-order estimates can be made which give good visual results in texture generation experiments explained later, we cannot reliably perform a chi-square test of them on the 64 X 64 lattice. The number of cells with only one member is very large since the number of theoretically possible cells with a fully anisotropic third-order model is 729, yet there is a maximum of 455 distinct configurations in a single third-order coding.

Our preferred choices for the best-fitting model are based on a simple rule: choose the model that gives the largest number of acceptances with a conservative decision rule. The results of this rule are displayed in Table III. We would not consider a fit to be adequate unless the majority of samples from the texture class fit the model.

2) *Binomial Texture Results:* With the experience gained from the binary fits, we limited our attention to four samples of size 128 X 128 from each of eight gray level pictures. First-

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TABLE IV
BEST FITTING EIGHT GRAY LEVEL TEXTURE MODELS. IN THE TABLE BELOW, THE "*I*" INDICATES THAT THE FEWEST REJECTIONS WERE OBTAINED BY USING ISOTROPIC ESTIMATION, WHILE "*A*" INDICATES THAT THE BEST RESULTS WERE OBTAINED USING ANISOTROPIC PARAMETERS. THE SYMBOL "—" INDICATES THAT NEITHER MODEL WAS APPROPRIATE.

Texture Name	First Order	Second Order	Acceptances at 5%
Bricks	A	I	3
Ceiling Tile	A	I	3
Cork	A	A	4
Fur	I	A	4
Grass	A	A	4
Paper	A	A	4
Pebbles	---	---	0
Sand	A	A	4
Screen	---	---	0
Water	---	---	0
Wood (1)	A	A	1
Wood (2)	---	---	0

order models yielded consistently negative results. The binomial model is not suitable for bimodal or uniform conditional probabilities because it always has a peak at exactly one value for any choice of θ. Also, if there are two likely gray values which are not contiguous, then no choice of the parameter θ can yield the correct probabilities.

As in the binary case, third-order analysis cannot be performed on samples of size 64 X 64 for eight gray level textures. Good results were obtained for all but the inhomogeneous textures: water, wood, and pebbles. The binomial model has difficulty in handling large areas of equal brightness. All of the textures which did not fit the model well are either blotchy or regular, like the image of the screen. Fine-grained textures can be handled and, as we shall see, generated by the binomial model easily. The best results of the two sets are shown in Table IV.

In both Tables III and IV, the interpretation of the acceptances column requires some clarification. Each texture is estimated four times (on each of the four codings) for a second-order model. The conservative decision rule at the 5 percent significance level says that if any of the significances of the chi-square values are under 0.05/4 = 0.0125, then we reject the texture at the 5 percent level as belonging to the texture class.

D. Texture Matching Experiments

This section examines the viability of the Markov random field as a supervised texture generation procedure. The input texture is measured using the maximum likelihood approach described in Section V-C. The results of that evaluation are used as input to the generation procedure in Fig. 2. As mentioned in Section V-A, the third-order model on a 64 X 64 texture cannot be reliably fitted since it causes too many empty or near-empty cells. Limited experimentation showed that if the image size was increased to 128 X 128, a sensible chi-square estimate could be made, although the parameters did not change very much from the estimates made on the 64 X 64 textures.

The steps performed in the texture matching experiments were as follows.

a) The parameters of a digitized natural texture were estimated using a model of some order.

b) If the texture was accepted as a Markov random field using the conservative hypothesis test, then an attempt was made to generate a synthetic texture using the estimated parameters of the natural texture.

c) The parameters of the synthetic texture were estimated in order to assure that the synthetic texture has the same (or very nearly) parameters as the original texture sample.

d) At this point, we have two textures with the same Markov random field parameters. The two textures can then be compared to see if they are visually similar.

1) *Binary Texture Matching:* The parameters used to generate the synthetic textures were obtained by averaging the parameter estimates from each of the codings of a single subimage. The choice of subimage was arbitrary. A third-order estimate was used in all cases except for the pictures of wood grain (1) and pebbles. In these two cases, a first-order model gave better visual results. First-order models tend to form in blob-like aggregations and cannot correctly characterize fine-structured textures. Moreover, they are inadequate in showing any directionality except vertical and horizontal.

The regularity and neat rectangles are not present in the synthetic texture of brick. The inhomogeneity of fur, wood(1), and wood(2) is missed in the synthetic examples. The remaining eight textures (cork, screen, sand, ceiling tile, grass, water, paper, and pebbles) are reasonably approximated by the simulated textures.

The clustering present in the pebbles image, Fig. 17(a), is shown clearly in the synthetic version, Fig. 17(a1). In the picture of cork, Fig. 17(b), diagonality is the overriding feature and this is correctly modeled. The screen image, Fig. 17(c1), is remarkably similar to the original, Fig. 17(c). The third-order repulsion effect provides the curious checkerboard effects along the lines in both the original and the generated texture.

2) *Binomial Texture Matching:* A 128 X 128 image size was used for both estimation and matching. The natural textures were quantized to have eight gray levels using histogram equalization. In all cases, a third-order model was estimated and used to generate the synthetic textures. Textures which failed in this matching experiment were: water, wood grain (1), wood grain (2), and pebbles. When considered as an eight-gray level image, these textures take on a distinctly inhomogeneous appearance. At this size, they look like pictures of objects, whereas the generated textures look like a fine-grain field. The Markov random field always results in a homogeneous covering of the image, which cannot be a blotchy image unless the parameters are extreme. As an example, the fur pictures, Fig. 18(a) and (a1), show the result of an inhomogeneity in the image.

The other clear failure is the bricks picture. As in the binary case, the bricks image has a regular structure. The Markov random field can only detect a hint of a vertical structure. Ceiling tile, cork, grass, paper, and sand are handled adequately. Missing are the large black holes in the synthetic tile picture, but there are some dense black patches [Fig. 18(c) and (c1)]. The distinctly three-dimensional appearance of the handmade paper, admittedly a tactile property is not captured either [Fig. 18(b) and (b1)].

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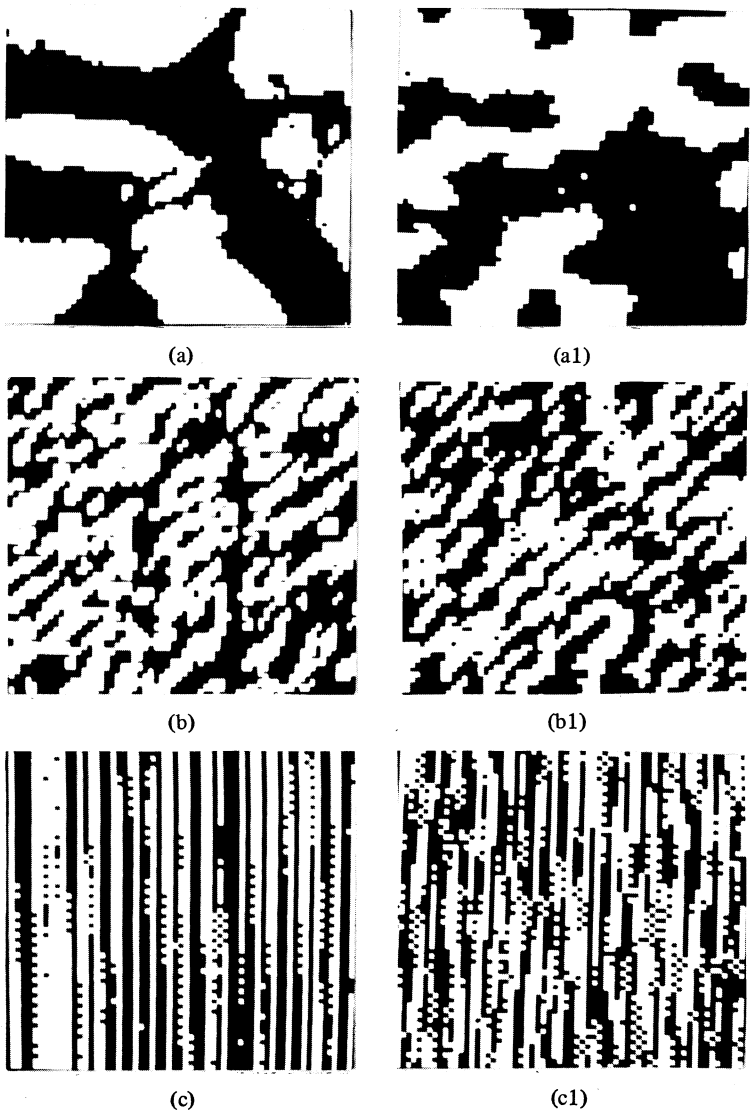


Fig. 17. Real and synthetic binary textures: (a) binary pebbles, (a1) synthetic binary pebbles, (b) binary cork, (b1) synthetic binary cork, (c) binary screen, (c1) synthetic binary screen.

The model seems to be adequate for duplicating the microtextures, but is incapable of handling strong regularity or cloud-like inhomogeneities. These experiments should be taken as an exploration of the limits of a purely statistical approach to texture without any *a priori* knowledge at all. For example, if we knew that the bricks picture was supposed to have rectangles of a certain size and orientation, then we could start with them as an outline and then fill in the rectangles with a Markov random field.

VI. Conclusions and Discussion

A. Summary

The focus of this research was on the application of the binomial model for texture, with special attention to the selection of an appropriate conditional distribution for the points of a lattice. It was demonstrated that the Markov random field parameters control the strength and direction of the clustering of the image. Overall, the microtextures, grass, sand, cork, ceiling tile, and paper, obeyed the Markov random field model well.

Using the estimation procedure for Markov random fields, the parameters of the natural textures were measured. The measured parameters were used as input to the texture generation procedure. This was an attempt to see how far the statistical approach alone could be carried in the absence of any structural information about the texture. Microtextures were successfully generated although the regular and inhomogeneous textures bore little resemblance to their synthetic counterparts.

The positive aspects of the Markov random field model are given as follows.

1) The model is fully two-dimensional and does not assume

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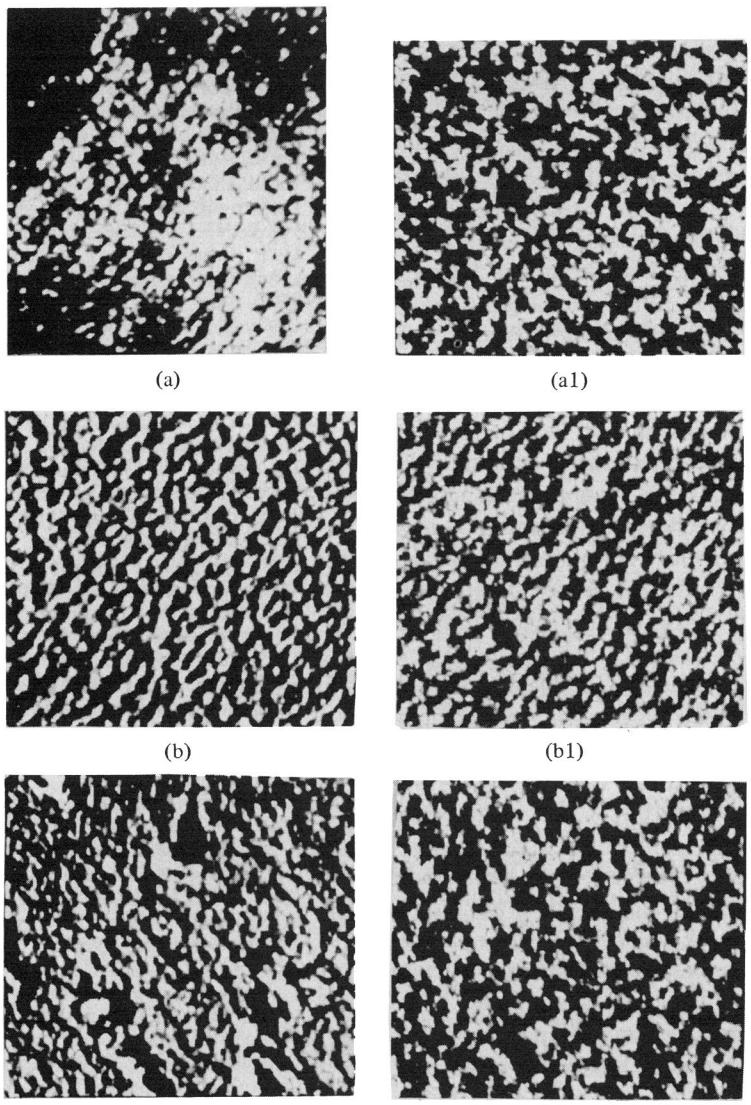


Fig. 18. Real and synthetic eight level textures. (a) fur, (a1) synthetic fur, (b) paper, (b1) synthetic paper, (c) ceiling tile, (c1) synthetic ceiling tile.

a causal (unilateral) dependence [11], [34]. This means that the Markov property is not relative to a particular direction. The Markov random field model allows us to consider neighbors in all directions.

2) The texture parameters are measurable from samples and the appropriateness of the model can be assessed objectively by a hypothesis test. Moreover, the Markov random field model allows the fitting of the sample texture directly to the model parameters. The current state of most models requires indirect fits by using variograms and correlation matches [15], [45].

3) The model parameters themselves are sufficient to generate images. In a purely feature-based approach, we do not have the capability to generate images from features because the features do not uniquely define a texture class.

4) The pattern formation process, although specified locally, implies a global pattern. The consistency conditions enforced by the Markov random field cause a pattern over the entire lattice. As Besag explains in the discussion of his lattice model paper, [4]:

> Incidentally, the fact that a scheme is formally described as "locally interactive" does not imply that the patterns it produces are local in nature (cf. the extreme case of long-range order in the Ising model).

5) The patterns formed are realistic. The binomial model allows natural smooth peaks and valleys in the gray level height field over the plane.

6) The patterns generated by varying the model parameters can be studied and classified. Directionality, coarseness, gray level distribution, and sharpness can all be controlled by choice of the parameters.

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7) The parameter estimation procedure can be implemented in a parallel algorithm. Each parameter estimate is performed on disjoint codings, each of which can be processed separately and then averaged to form the final result. Even the hypothesis tests could be done in parallel.

The Markov random field model has the following disadvantages.

1) Regular textures are not modeled very well.

2) The textural primitives of the Markov random field are nongeometric.

3) Large images are required to get good parameter estimates.

4) The theoretical properties of the model are, in general, difficult to obtain. We are currently computing the co-occurrence matrices as a function of the Markov random field parameters.

Several additional experiments can be done to determine the limitations and power of the Markov random field models. The full second-order model involving nonlinear terms may provide sufficient information about the image without any need to go as far as third-order. The full second-order model has only four codings, which is a distinct computational advantage.

The Brodatz texture samples certainly do not exhaust the possibilities of the model. It may be possible, for example, to use the Markov random field as a method of modeling film grain noise [23]. The binomial model textures have a distinctly biomedical flavor and may be of use in modeling images obtained by microscopy.

The Markov random field model can be used in a hierarchical manner. First, estimate the parameters on the given resolution. Next, consider the lattice of dimension half the original dimension by either averaging nonoverlapping two by two neighborhoods or simply selecting every other point. The reduced lattice parameters are then estimated and the process is repeated to get a sequence of parameter vectors which may describe the structure of the image better than a single set. We are investigating the use of Markov random field parameters for texture discrimination.

A test of homogeneity is needed. The Markov random field model assumes homogeneity in the image. Some verification of this should be made before estimating the parameters and assessing the goodness-of-fit.

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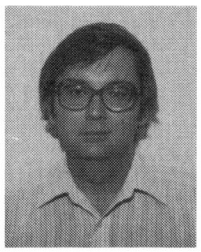
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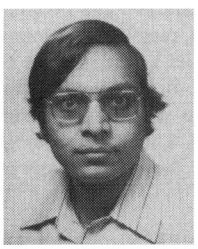
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